MILITARY SPECIFICATION

PROTECTOR, STATIC TUBE: H4049A

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of cover for a self-aligning static tube, entitled Protector Static Tube H4049A, which is used on the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8861168	- Protector, Static Tube H4049A.
EL-8869336	- Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The cover shall comply with all requirements specified on the drawing (dwg) 8861168, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Humidity. The cover shall withstand exposure to the humidity cycles specified in MIL-E-5272, paragraph 4.4, procedure I, without evidence of deterioration such as cracking, peeling, splitting or other damage when tested as specified in 4.4.1.

3.5 Welding. Welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of fusion, penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of skill or experience on the part of the operator. All welding flux, scale, weld spatter, acid and basic solutions shall be completely removed prior to applications of the finish coatings.

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8861168) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Humidity (see 3.4)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Body section weldment assembly (see dwg 8861151 and 8861158 covering a detail of dwg 8861168.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Flatness at edge of cylinder half, incorrect	Gage

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102. Weld inadequate, missing or improper (see 3.5).....Visual

Minor: None defined.

4.3.2.1.2 Protector, static tube sub-assembly (see dwg 8861163 covering a detail of dwg 8861168).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Gap between body sections including parallelism of body section edges when fasteners are locked	Gage
*102.	Misalignment of two halves at mating edges	Visual
103.	Rivet loose, cracked or missing	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.6)	Visual
202.	Protective finish inadequate, including bare spot	Visual
203.	Hardware loose or missing	Visual/Manual
204.	Rigidity of hinge	Manual

*If the halves appear to be misaligned a gage (see 4.3.4.1) shall be used to determine the extent of misalignment. If the extent of misalignment exceeds the tolerance specified on the drawing, the assembly shall be classified defective until corrective action has been taken.

4.3.2.1.3 Protector, static tube H4049A (see dwg 8861168).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Marking missing or unidentifiable	Visual
102.	Gaskets missing, damaged or loose	Visual/Manual
103.	Latches fail to function properly	Manual
104.	Screw, guide pin functions inadequately	Manual

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Minor:

201. Evidence of poor workmanship (see 3.6).....Visual
202. Adhesive fails to hold cone securely.....Manual

4.3.2.1.4 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.6)	Visual

4.3.3 Testing. No tests are applicable to the cover.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8869336). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Humidity. The cover shall be mounted at an angle of 45 degrees, plus or minus 5 degrees, in a humidity test chamber and shall be subjected to the humidity test cycles specified in MIL-E-5272, paragraph 4.4, procedure I. All temperatures shall be maintained at + 5°F. The pH value of the water used shall be adjusted by the use of chemically pure hydrochloric acid or chemically pure sodium hydroxide. Upon completion of the last cycle, the cover shall be immediately removed from the chamber. Excessive moisture may be removed by blowing dry, filtered air over the unit for a period of 5 minutes.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cover shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169 (MU).

5.1.2 Level C. The cover shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169 (MU).

5.2 Packing.

5.2.1 Levels A and C. The cover shall be packed in accordance with the applicable instructions contained in MIL-STD-1169 (MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169 (MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-2355 dated 29 March 1962.

Custodian:
Army - PA

Preparing activity:
Army - PA

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